

GUO ZHIHAN

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09/2022 - 06/2026 Southwest University(SWU) Bachelor of Engineering in Software Engineering

GPA:83.086/100 IETES:6.0

Class Library and Data Structure:91 Information Security Principle:87 Advanced Mathematics IB:84 Artificial Intelligence and Industrial Application:91 Comprehensive Practice of Data Structure and Algorithm:94

PUBLICATION

First Author, Comprehensive Analysis of Fruit Detection, Ripeness Assessment, and Mass Estimation Using YOLOv7, ResNet, and ViT Models, 2024 IEEE 3rd International Conference on Electrical Engineering, Big Data and Algorithms, 2024.2, 1647

HONOR AND AWARD

Third-class scholarship of Southwest University (top 17.5%)	10/2024
Silver Award in the 26th CRAIC - New Quality Productivity Innovation Design Special Competition	05/2024
Bronze Award in the 26th CRAIC - Artificial Intelligence Innovation Competition	04/2024
Bronze Award in the 15th Lanqiao Cup Digital Technology Innovation Competition	04/2024
Silver Award in the 14th China National Undergraduate “Innovation, Creativity and Entrepreneurship”	04/2024
Honorable Mention Award in Mathematical Contest In Modeling	02/2024
Meritorious Award in the 12th Certificate Authority Cup International Mathematical Contest in Modeling	11/2023
Silver Award in Gauss Cup National College Student Mathematical Modeling Challenge	11/2023
Silver Award in the 18th National University Students Intelligence Car Race	11/2023

PROJECT EXPERIENCE

Generation of Wargame Sand Tables Based on Diffusion Models and Text-Driven Techniques

Member 01/2024 - 02/2025

- Integration of diffusion modelling and text-driven technology to develop a military chess sandbox generation system designed to provide players with an immersive military strategy gaming experience;
- Comprehensively collect and clean and annotate the image data of the military chess sand table, enrich the dataset through data enhancement techniques such as rotation and scaling, and improve the model representation and generation quality;
- The speed, quality and clarity of model generation are improved multidimensionally by hyperparameter tuning, network structure optimisation, regularisation and other techniques to effectively prevent overfitting;
- Comprehensively assessed the performance of the model, further optimised the model based on subjective evaluations by military experts and military chess deduction enthusiasts and wrote a 10,550-word project declaration, which was ultimately approved as a national project.

“Cun Buyixing”——Intelligent Unmanned Delivery Vehicle for Complex Scenarios

Member 11/2023 - 03/2024

- Utilized vertical rudder wheels and NXT silane as coupling agents to optimize hardware material selection for the cart and equipped with multimodal sensors for comprehensive information perception.
- Designed an innovative wheel-track composite structure to overcome limitations of various terrains such as slopes, stairs, and elevators and expanded the delivery range to solve the ‘last centimeter’ challenge.
- Introduced a semantics-driven model for in-depth analysis of text commands to enhance cart autonomy and intelligence, designed an intelligent control unit incorporating gyroscope and hydraulic technology to ensure vehicle stability.
- Implemented GPS, BeiDou Navigation Satellite System and so on for high-precision positioning and route navigation, employed RVIZ tools for real-time three-dimensional visualization tracking of the unmanned delivery vehicle’s motion status and travel routes.
- Integrated TOF radar and Moxley laser navigation technologies for obstacle avoidance and path exploration, drew a 3D diagram of the cart, and created an operational demonstration video with CAD.
- Wrote a 78-page report and instruction manual and secured a provincial-level project approval.

Research on Apple Detection and Quality Analysis System Based on Multi-Model Fusion

Program Leader

11/2023 - 2/2024

- Target detection and counting: YOLOv7 model was used to achieve apple target detection, 200 samples were detected using SPSS, and the number of apples in a single image was counted by the number of detection frames, and a histogram of the distribution of the number of apples generated was obtained.
- Maturity grading assessment: Definition of three stages of maturity criteria, calculation of maturity based on red/yellow/green pigment weights, and development of a HSV colour space analysis model.
- Mass estimation algorithm: Introduce the coin reference to establish the pixel-mass conversion model, derive the mass calculation formula, then estimate the mass of a single fruit by the red pixel area.
- Comparative Analysis of Classification Models: Constructing ResNet and ViT Dual Model Classification Frameworks, Validation Set Accuracy Comparison Experiments.

INTERNSHIP EXPERIENCE

Zhejiang Huawei Communication Technology Co., Ltd.

03/2025 - 04/2025

Technical Training engineer, ICT BG - Guian Delivery Team

- Systematically studied the fundamentals of machine learning and deep learning, mastered the principles and application scenarios of core algorithms such as linear regression, decision trees, convolutional neural networks (CNNs), and recurrent neural networks (RNNs), and completed related practical exercises and certification exams.
- Responsible for the development of image recognition and speech-to-text modules for the digital human broadcasting project. In image recognition tasks, mastered CNN model construction and data preprocessing techniques; during speech recognition development, further studied speech signal feature extraction (such as MFCC) and end-to-end deep learning model construction, and improved model recognition accuracy through practical application.
- Mastered methods for deploying and optimising AI system engineering, which includes using Docker containers for unified environment deployment, effectively resolving dependency library conflicts and significantly improving system stability and operational efficiency.

Guizhou Zhicheng Technology Co., Ltd.

07/2024

Big Data Analytics Engineer, System Research and Development Department

- Utilized technologies such as Hadoop and Spark to mine bus station data and card swipe data for accurately determining passenger boarding and alighting stations.
- Participated in the design of an algorithm based on transfer coefficients, achieved automatic identification and analysis of passenger transfer behaviors between different routes, and calculated the average transfer coefficient.
- Built a time series model to forecast the volume of passenger travel during different periods.
- Employed tools like Tableau and Matlab for data visualization and enhanced the ease of data interpretation and the effectiveness of decision support.

EXCHANGE EXPERIENCE

Hong Kong University

Leader

06/2025

- Learned how to use generative AI to generate images and videos. Then used generative AI in education to assist people in learning Chinese idioms.
- Learned about the Python programs on Colab and how to use prompts to enhance the quality of AI-generated images and videos.
- Analyzed the impact of different prompts on the AI and compared the differences between different AI platforms.
- Accomplished all the workshop and delivered a 15-minute presentation in English.

The Chinese University of Hong Kong

Member

02/2024

- Conducted a research project on a Blockchain-based Food Supply Chain Maintenance System, analyzed the cost structure, traceability system, key operations, and revenue sources of the food supply chain at Qian Da Ma Fresh Supermarket.
- Compared successful cases like Walmart and Carrefour and proposed strategic transformations for integrating blockchain into the food supply chain business model.
- Delivered a 34-minute presentation in English and achieved a grade of A- (88-92).

STUDENT WORK

Xingke Debate Team, College of Computer and Information Science, SWU

Member

09/2022 - 07/2024

- Participated in various school-level debate competitions.
- Took responsibility for planning, on-site hosting, and communication coordination of debate competitions.